

Biology

- 1) Both ATP and NADPH are produced in the light reaction and are then utilised in the dark reaction. Hence, they are the substances which connect ATP and NADPH.
- 2) It occurs in mitochondria.
- 3) It is devoid of red blood cells.
- 4) Plants get rid of water through the process of transpiration.
- 5) The glomerulus and Bowman's capsule together form Malpighian body.

8).	Light Reaction	Dark Reaction
	There is no sugar formed	Sugar is formed.
	Oxygen is released	Oxygen is not released.
	occurs in grana region.	occurs in stroma region.

Date _____

9) a) Pepsin in Stomach act on proteins.
~~Pepsinogen~~
proteins $\xrightarrow{\text{pepsin}}$ peptides.

Later trypsin released by pancreas helps to digest protein. Protein is finally broken down to amino acids ~~in~~ by intestinal juices.

b) The pyruvate gets converted into lactic acid in the absence of oxygen in cytoplasm

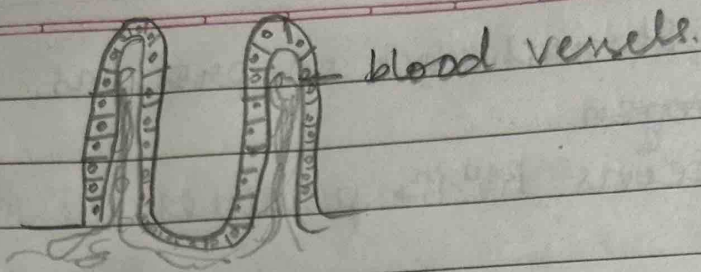
Glucose $\xrightarrow{\text{cytoplasm}}$ Pyruvate $\xrightarrow[\text{No } O_2]{\text{cytoplasm}}$ Lactic acid + 2 ATP

10) a) Acid released is HCl. It is used to kill the germs in the food.

• It also activates two enzymes.

Pepsinogen $\xrightarrow{\text{HCl}}$ Pepsin
Prorennin $\xrightarrow{\text{HCl}}$ Rennin

b)



Lacteal helps in transportation of absorbed food especially fats. It absorbs fat in the form of fatty acids and glycerol.

11 a) The blood capillaries in the lungs increases ^{efficient} rate of exchange of gases as it provides ~~and~~ a larger surface area and constitute thin walls.

b) These cartilaginous rings prevent the collapsing of trachea even when there is not much air in it.

12 a) During winter the pressure in kidney increases cause more fluid to filter out. Moreover, as the temperature is low there is less sweating, thus no evaporation losses. Due to these reasons, ultimately more urine is produced.

b) Lungs, skin and liver are

the accessory excretory organs

c) Malpighian tubule is the excretory organ.

(13) i) 4 - Right atrium
5 - left atrium

(ii) 3 - superior vena cava

- helps in bringing deoxygenated blood to the heart from upper body

- 6 - inferior vena cava ~~helps to~~
 - helps in bringing deoxygenated blood to the heart from lower part of body

(iii) Aorta carries blood away from heart

(iv) The blood passes twice through the heart in one cycle. Once through the heart and lungs (pulmonary circulation) and the other through heart and various body parts (double circulation).

7) In case of plants, respiration occurs through diffusion via lenticels, roots and leaves. In animals, Respiration occurs generally in lungs and they show cellular respiration